

Linear Regression and Baselines

Estimate continuous outcomes with interpretable baselines and robust evaluation.



Learning outcomes

01 ——— Establish naive and linear baselines.

02 ——— Interpret coefficients within assumptions.

03 ——— Diagnose residual patterns.

The concept system

01

baseline

02

least squares

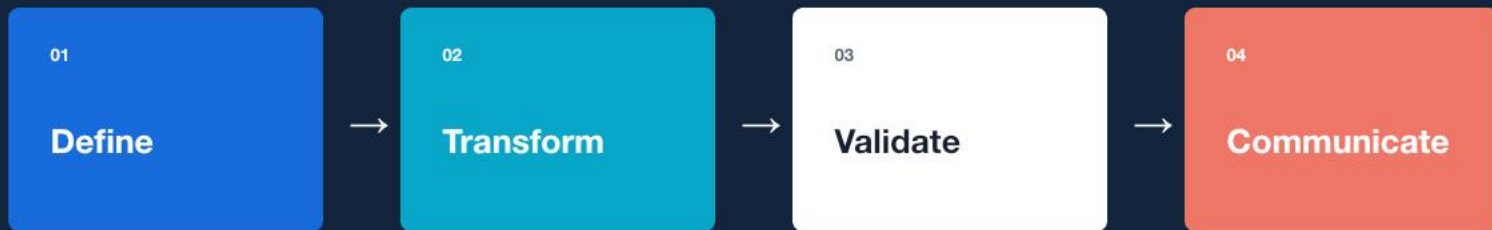
03

coefficients

04

residuals

A repeatable reasoning loop



Estimate delivery time and compare against a median baseline.

Guided lab

LAB BRIEF

Estimate delivery time and compare against a median baseline.

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A baseline report with residual diagnostics.

QUALITY GATE

-  **Reproducible**
-  **Tested**
-  **Decision-relevant**

WEEKLY ASSESSMENT

Regression Model Card

Evaluation 30%, diagnostics 25%, model choice 20%, reproducibility 15%, communication 10%.

What should remain after the lesson?

CONCEPT

baseline + least squares

PRACTICE

Estimate delivery time and compare against a median baseline.

EVIDENCE

A baseline report with residual diagnostics.
